

Windows Technical Troubleshooting Guide

FIELD REFERENCE & SYSTEMS OPTIMIZATION HANDBOOK

Author Corporation: NCB Technical Solutions | **Purpose:** System Diagnostic, Recovery, and Deployment Workflows.

1. View All Installed Applications

Shortcut: Win + R

Target: Run Dialog

Launches a specialized system shell folder displaying every single application installed on the operating system, including hidden system apps.

```
shell:AppsFolder
```

2. Fixing System Slowness & Corrupted Files

Access: Command Prompt (CMD)

Privilege: Run as Administrator

Executes the System File Checker (SFC) tool to scan, verify, and automatically repair corrupted or missing critical Windows system files.

```
sfc /scannow
```

3. Clearing Temporary Files & Storage

Access: CMD / Run Dialog

Privilege: Run as Administrator

Opens the native Windows Disk Cleanup utility configuration to rapidly purge temporary files, cache, and system logs to free up storage space.

```
cleanmgr
```

4. Fixing Repeated Freezing & Lagging (Disk Check)

Access: Command Prompt (CMD)

Privilege: Run as Administrator

Invokes the Check Disk utility to scan the filesystem and hard drive sectors for physical or logical errors, fixing integrity issues directly.

```
chkdsk /f /r
```

Pro Tip: If the volume is currently in use, the system will ask to schedule the check on the next reboot. Type Y, hit Enter, and restart the PC.

5. Troubleshooting Driver Issues (Wi-Fi, Bluetooth, Hardware)

Shortcut: Win + R

Target: Run Dialog

Forces execution of the Microsoft Support Diagnostic Tool tailored for analyzing hardware configurations and repairing faulty core device drivers.

```
msdt.exe -id DeviceDiagnostic
```

6. Bulk Updating All Software

Access: Command Prompt (CMD)

Privilege: Run as Administrator

Leverages the Windows Package Manager (Winget) to evaluate all installed third-party programs and update them to their newest stable versions concurrently.

Check for available updates:

```
winget upgrade
```

Execute automated silent upgrade for all software:

```
winget upgrade --all
```

7. Advanced System Image Repair (DISM)

Access: Command Prompt (CMD)

Privilege: Run as Administrator

Utilizes the Deployment Image Servicing and Management (DISM) engine to verify the baseline health of the local Windows OS image component store. Run this if SFC fails.

```
DISM /Online /Cleanup-Image /CheckHealth
```

Deep Repair: To actively pull clean files down via Windows Update to overwrite corrupted OS components, run: `DISM /Online /Cleanup-Image /RestoreHealth`

8. Built-in Windows Malware Scanning

Shortcut: Win + R

Target: Run Dialog

Triggers the standalone Microsoft Malicious Software Removal Tool (MRT), allowing deep scanning of system directories independent of standard active antivirus agents.

```
mrt
```

9. Windows Password Reset Bypass (Utilman Accessibility Method)

A critical emergency recovery technique to regain administrative access to local user profiles by temporarily remapping the system Accessibility button to an elevated Command Prompt container.

Step 1: Boot to Recovery Command Prompt

1. At the primary Windows OS login interface, press and hold down the physical **Shift** key while clicking **Restart**.
2. Navigate precisely through the menu options: **Troubleshoot** → **Advanced options** → **Command Prompt**.

Step 2: Swap the Utility Manager Executable with CMD

Identify your primary operating system drive volume (typically C :) and swap binary identities using file renames:

```
C:
cd windows
cd system32
ren utilman.exe utilman1.exe
ren cmd.exe utilman.exe
```

Exit the command container and click **Continue** to reload the primary system login portal.

Step 3: Access SYSTEM Command Prompt and Reset Account

1. At the login screen, click the ****Accessibility**** icon (the human shape element situated at the lower right corner).
2. An elevated command shell window will deploy under SYSTEM-level authorization.
3. Open the Graphical User Account management module immediately:

```
control userpasswords2
```

Alternative Method: Reset credentials immediately via pure command line input:

```
net user [username] [new_password]
```